CYBEREXPLAIN AI

Explainability and Transparency in Multi-Agent Cybersecurity Decision-Making

Technical Architecture, Implementation, and Full Source Code Report

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GitHub Repository: https://github.com/umar-khatri/cyber-explain-agent
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Status: Production Ready & Tested
Backend: Python 3.13, FastAPI, LangGraph, Groq (openai/gpt-oss-120b)
Database: Supabase PostgreSQL
Frontend: Next.js 16 (App Router), TypeScript, Tailwind CSS, React Flow

Executive Summary:
CyberExplain AI demonstrates how collaborative multi-agent architectures enhance accuracy, eliminate ungrounded hallucination, and enforce explainability in high-stakes cybersecurity incident response. By orchestrating specialized agents (Threat, Risk, and Verification) via LangGraph, the system identifies unsupported claims, detects contradictions, and dynamically revises assessments before final security recommendations are synthesized.

1. System Overview & Problem Statement

Modern Security Operations Centers (SOCs) are inundated with alerts and speculative analyst summaries. Conventional large language model (LLM) implementations suffer from confirmation bias and hallucinated certainty: when presented with an incident containing panic-induced assertions (e.g., 'the server is fully compromised and all customer data is exfiltrated'), a single-pass model frequently accepts speculative claims as ground truth, elevating severity to Critical and prompting costly, unnecessary incident escalations.

CyberExplain AI resolves this vulnerability through multi-agent adversarial verification:

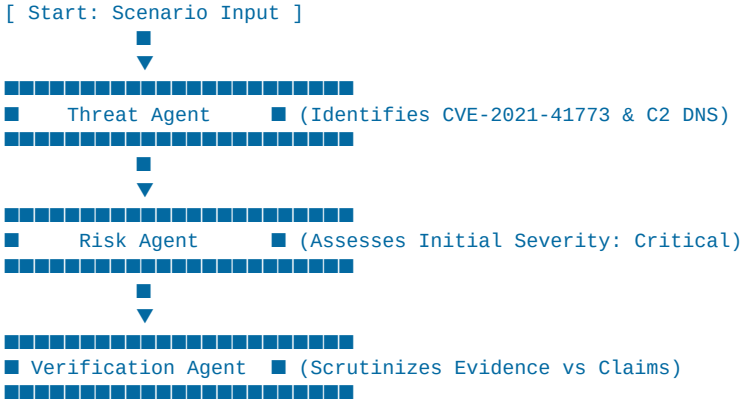
- **Independent Threat Formulation:** The Threat Agent catalogues observables (e.g., Apache 2.4.49, /cgi-bin/ requests, C2 DNS queries) while tagging evidence-based indicators versus speculative interpretations.
- **Severity Modeling:** The Risk Agent evaluates exploitability, impact, and provisional severity (initially Critical due to alleged data loss).
- **Adversarial Verification:** The Verification Agent cross-references every claim directly against the raw incident telemetry, flagging unsupported assertions, contradictions, and data gaps.
- **Conditional Revision:** LangGraph routes the state back through Threat and Risk agents to correct claims, downgrade unverified severities, and produce an evidence-backed Final Assessment.

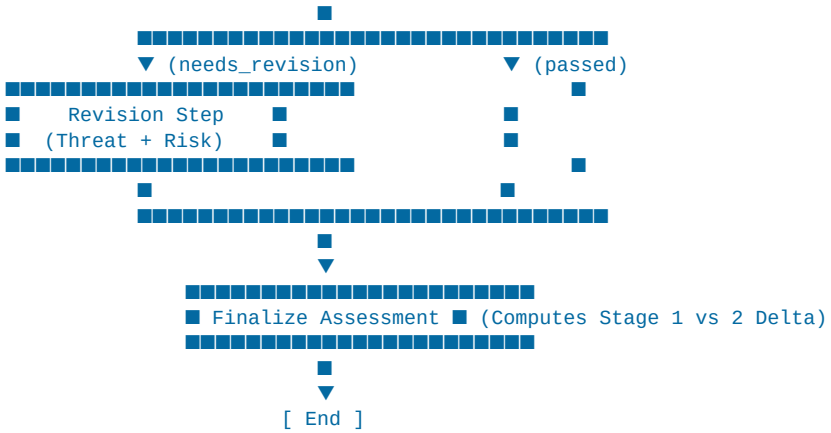
2. System Architecture & Workflow

Agent / Component	Role & Responsibility	Input Source	Output Artifact
Threat Agent (Stage 1)	Extracts CVEs, malicious IPs, DNS domains, and access patterns.	Raw Scenario Text	Threats list, evidence quotes, model interpretations.
Risk Agent (Stage 1)	Calculates initial risk score (0-10) and severity rating.	Threat Analysis + Scenario	Severity, risk score, preliminary recommendations.
Verification Agent	Critiques claims, isolates contradictions, flags unsupported assertions.	Scenario + Threats + Risk	Supported/Unsupported claims, requested corrections.
Revision Loop (Stage 2)	Revises threats, lowers confidence, recalibrates severity.	Verification Feedback	Revised threats and downgraded realistic risk score.
Final Synthesis Node	Generates executive report and Before/After analysis.	All Agent State	Consolidated assessment with transparent explainabilit
Supabase PostgreSQL	Persists assessment runs, node telemetry, and graph edges.	FastAPI Assessment Router	Persistent UUIDs, timestamps, and queryable audit trai

3. Multi-Agent Decision State Graph (LangGraph)

The multi-agent system is compiled into a cyclic StateGraph using LangGraph. Unlike linear chains, LangGraph allows conditional branches where the graph autonomously decides whether to route to revision or straight to finalize_assessment based on the Verification Agent's overall_status.




(needs_revision)
(passed)
Revision Step
(Threat + Risk)
Finalize Assessment
(Computes Stage 1 vs 2 Delta)
[End]

4. Deep Dive: Explainability & The Benchmark Scenario

The Benchmark Cybersecurity Scenario:
"Our company's web application server (running Apache 2.4.49) was accessed from an unusual IP address (203.0.113.42) outside of business hours (3:00 AM local time). The server logs show multiple requests to /cgi-bin/ directories. Additionally, our internal DNS server has been resolving queries for known malicious domains (e.g., malware-c2.evil.example, data-exfil.bad.example). A junior analyst reviewed the logs and claims that the server has already been fully compromised and that attackers have exfiltrated all customer data, based solely on the unusual access pattern. No forensic analysis or memory dump has been performed yet."

How Explainability Solves the Assessment Problem:

Phase	Severity	Risk Score	Confidence	Key Finding & Evidence Grounding
Stage 1 (Initial)	Critical	9.0 / 10	0.65	Analyst claim of mass customer data exfiltration is taken at face value. Alarming panic
Verification	Needs Revision	—	—	Identifies that claim 'Server fully compromised and all data exfiltrated' has ZERO supp
Stage 2 (Revised)	High	7.6 / 10	0.88	Exfiltration downgraded to 'Unverified Hypothesis' with 0.18 confidence. Active Apach

5. Database Schema & REST API Specifications

PostgreSQL Persistence (Supabase): Assessments and complete execution traces are stored in the assessments table. JSONB columns allow rapid serialization of dynamic agent outputs, graph edges, and per-step timing.

Method	Endpoint	Description	Payload / Return
GET	/health	System health & credentials status	<code>{'status': 'ok', 'groq_configured': bool, ...}</code>
GET	/api/sample-scenario	Retrieves benchmark cyber incident	<code>{'scenario': string}</code>
POST	/api/assess	Executes full multi-agent LangGraph pipeline	Body: <code>ScenarioRequest</code> → <code>Assessm
GET	/api/assessments	Lists recent 50 assessments	<code>{'assessments': list[AssessmentListItem]}</code>
GET	/api/assessments/{id}	Fetches detailed assessment & full trace	Returns complete <code>AssessmentResponse</code>

6. Complete Source Code Listings

This section contains the full, verified source code files implementing the backend agents, orchestration graph, data models, database integrations, API endpoints, schema definitions, and frontend components.

File: Database Schema (Supabase SQL) ***(backend/schema.sql)***

```
-- CyberExplain AI: Supabase Schema
-- Run this in the Supabase SQL Editor to create the required tables.

CREATE TABLE IF NOT EXISTS assessments (
    id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    scenario TEXT NOT NULL,
    initial_threat_analysis JSONB,
    initial_risk_assessment JSONB,
    verification_result JSONB,
    revised_threat_analysis JSONB,
    revised_risk_assessment JSONB,
    final_assessment JSONB,
    execution_trace JSONB DEFAULT '[]'::jsonb,
    graph_edges JSONB DEFAULT '[]'::jsonb,
    created_at TIMESTAMPTZ DEFAULT NOW()
);

-- Index for faster lookups
CREATE INDEX IF NOT EXISTS idx_assessments_created_at ON assessments (created_at DESC);

-- Enable Row Level Security (optional, disable for dev)
-- ALTER TABLE assessments ENABLE ROW LEVEL SECURITY;
-- CREATE POLICY "Allow all" ON assessments FOR ALL USING (true);

```

File: Configuration & LLM Settings ***(backend/app/config.py)***

```
"""Application configuration loaded from environment variables."""

from pydantic_settings import BaseSettings
from functools import lru_cache


class Settings(BaseSettings):
    """Application settings sourced from .env file."""

    groq_api_key: str = ""
    groq_model_name: str = "openai/gpt-oss-120b"
    supabase_url: str = ""
    supabase_key: str = ""
    host: str = "0.0.0.0"
    port: int = 8000

    model_config = {"env_file": ".env", "env_file_encoding": "utf-8"}


@lru_cache
def get_settings() -> Settings:
    return Settings()

```

File: Pydantic Data Models & Types ***(backend/app/models.py)***

```
"""Pydantic models for API requests, responses, and internal data structures."""

from pydantic import BaseModel, Field
from typing import Optional
from datetime import datetime


# ---------- Request models ----------

class ScenarioRequest(BaseModel):
    """User-submitted cybersecurity scenario."""
    scenario: str = Field(..., min_length=10, description="The cybersecurity scenario to analyse")


# ---------- Agent output models ----------

class ThreatItem(BaseModel):
    id: str
    name: str
    description: str
    indicators: list[str] = []
    evidence_from_scenario: str = ""
    is_evidence_based: bool = True
    confidence: float = Field(0.5, ge=0.0, le=1.0)


class ThreatAnalysis(BaseModel):
    model_config = {"protected_namespaces": ()}
    threats: list[ThreatItem] = []
    summary: str = ""
    overall_confidence: float = Field(0.5, ge=0.0, le=1.0)
    evidence_references: list[str] = []
    model_interpretations: list[str] = []


class RiskAssessment(BaseModel):
    model_config = {"protected_namespaces": ()}
    severity: str = "Medium"  # Critical, High, Medium, Low
    risk_score: float = Field(5.0, ge=0.0, le=10.0)
    evidence: list[str] = []
    reasoning: str = ""
    confidence: float = Field(0.5, ge=0.0, le=1.0)
    model_interpretations: list[str] = []
    recommendations: list[str] = []

```

```
|
| class ClaimCheck(BaseModel):
|     claim: str
|     status: str  # "supported", "unsupported", "contradicted"
|     evidence: str = ""
|     explanation: str = ""
|
|
| class VerificationResult(BaseModel):
|     supported_claims: list[ClaimCheck] = []
|     unsupported_claims: list[ClaimCheck] = []
|     contradictions: list[ClaimCheck] = []
|     corrections_requested: list[str] = []
|     overall_status: str = "pending"
|     explanation: str = ""
|
|
| # ---------- Execution trace ----------
|
| class TraceStep(BaseModel):
|     agent_name: str
|     step_order: int
|     stage: str  # "initial" or "revision"
|     input_summary: str
|     output_summary: str
|     timestamp: str
|     duration_ms: float = 0.0
|
|
| class GraphEdge(BaseModel):
|     source: str
|     target: str
|     label: str = ""
|     data_passed: str = ""
|
|
| # ---------- Full assessment response ----------
|
| class AssessmentResponse(BaseModel):
|     id: str
|     scenario: str
|     initial_threat_analysis: Optional[dict] = None
|     initial_risk_assessment: Optional[dict] = None
|     verification_result: Optional[dict] = None
|     revised_threat_analysis: Optional[dict] = None
```

```
|     revised_risk_assessment: Optional[dict] = None
|     final_assessment: Optional[dict] = None
|     execution_trace: list[dict] = []
|     graph_edges: list[dict] = []
|     created_at: str = ""
|
|
| class AssessmentListItem(BaseModel):
|     id: str
|     scenario: str
|     severity: str = ""
|     created_at: str = ""
|
```

File: Supabase Client & In-Memory Store ***(backend/app/database.py)***

```python
"""Supabase database client and helper functions."""

import json
from typing import Optional
from supabase import create_client, Client
from app.config import get_settings


# Local in-memory fallback cache for when Supabase is not configured
_MEMORY_STORE: dict[str, dict] = {}


def get_supabase_client() -> Optional[Client]:
    """Create and return a Supabase client if configured."""
    settings = get_settings()
    if not settings.supabase_url or not settings.supabase_key or "your-project" in settings.supabase_url:
        return None
    try:
        return create_client(settings.supabase_url, settings.supabase_key)
    except Exception:
        return None


async def store_assessment(assessment: dict) -> dict:
    """Store a completed assessment in Supabase or memory."""
    # Always keep in local memory store as cache
    _MEMORY_STORE[assessment["id"]] = assessment

    client = get_supabase_client()
    if client:
        result = client.table("assessments").insert(assessment).execute()
        return result.data[0] if result.data else assessment
    return assessment


async def get_assessment(assessment_id: str) -> Optional[dict]:
    """Retrieve a single assessment by ID."""
    if assessment_id in _MEMORY_STORE:
        return _MEMORY_STORE[assessment_id]

    client = get_supabase_client()
    if client:
        result = (
            client.table("assessments")
            .select("*")
            .eq("id", assessment_id)
            .execute()
        )
        return result.data[0] if result.data else None
    return None


async def get_all_assessments() -> list[dict]:
    """List all assessments ordered by creation time."""
    client = get_supabase_client()
    if client:
        result = (
            client.table("assessments")
            .select("id, scenario, final_assessment, created_at")
            .order("created_at", desc=True)
            .limit(50)
            .execute()
        )
        if result.data:
            return result.data

    # Fallback to local memory cache
    return [
        {
            "id": v["id"],
            "scenario": v.get("scenario", ""),
            "final_assessment": v.get("final_assessment", {}),
            "created_at": v.get("created_at", ""),
        }
        for v in reversed(list(_MEMORY_STORE.values()))
    ]

```

File: FastAPI Application Entry Point *(backend/app/main.py)*

```
"""FastAPI application entry point."""

import logging
from fastapi import FastAPI
from fastapi.middleware.cors import CORSMiddleware
from app.routes.assessment import router as assessment_router
from app.config import get_settings

# Configure logging
logging.basicConfig(
    level=logging.INFO,
    format="%(asctime)s ■ %(name)-25s ■ %(levelname)-7s ■ %(message)s",
    datefmt="%H:%M:%S",
)

app = FastAPI(
    title="CyberExplain AI",
    description="Multi-Agent Cybersecurity Risk Assessment with Explainability",
    version="1.0.0",
)

# CORS — allow the Next.js frontend
app.add_middleware(
    CORSMiddleware,
    allow_origins=["http://localhost:3000", "http://127.0.0.1:3000"],
    allow_credentials=True,
    allow_methods=["*"],
    allow_headers=["*"],
)

# Register routes
app.include_router(assessment_router)


@app.get("/")
async def root():
    return {
        "name": "CyberExplain AI",
        "version": "1.0.0",
        "docs": "/docs",
    }


@app.get("/health")
async def health():
```

```
    settings = get_settings()
    return {
        "status": "ok",
        "groq_configured": bool(settings.groq_api_key),
        "supabase_configured": bool(settings.supabase_url and settings.supabase_key),
    }


if __name__ == "__main__":
    import uvicorn
    settings = get_settings()
    uvicorn.run(
        "app.main:app",
        host=settings.host,
        port=settings.port,
        reload=True,
    )

```

File: Assessment & Pipeline Routes ***(backend/app/routes/assessment.py)***

```
"""Assessment API routes — submit scenarios and retrieve results."""

import logging
from fastapi import APIRouter, HTTPException
from app.models import ScenarioRequest, AssessmentResponse
from app.agents.graph import run_assessment_pipeline
from app.database import store_assessment, get_assessment, get_all_assessments

logger = logging.getLogger(__name__)
router = APIRouter(prefix="/api", tags=["assessment"])

# ■■ Sample scenario ■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■

SAMPLE_SCENARIO = """Our company's web application server (running Apache 2.4.49) was accessed from an \
unusual IP address (203.0.113.42) outside of business hours (3:00 AM local time). \
The server logs show multiple requests to /cgi-bin/ directories. \
Additionally, our internal DNS server has been resolving queries for known malicious domains \
(e.g., malware-c2.evil.example, data-exfil.bad.example). \
A junior analyst reviewed the logs and claims that the server has already been fully compromised \
and that attackers have exfiltrated all customer data, based solely on the unusual access pattern. \
No forensic analysis or memory dump has been performed yet."""


@router.get("/sample-scenario")
async def get_sample_scenario():
    """Return the built-in sample scenario."""
    return {"scenario": SAMPLE_SCENARIO}


@router.post("/assess", response_model=AssessmentResponse)
async def run_assessment(request: ScenarioRequest):
    """Submit a cybersecurity scenario for multi-agent assessment.

    This triggers the full pipeline:
    1. Threat Agent → identifies threats
    2. Risk Agent → assesses severity
    3. Verification Agent → checks claims
    4. (Conditional) Revision → corrects unsupported claims
    5. Final Assessment → combined result
    """
    logger.info("Starting assessment for scenario (%d chars)", len(request.scenario))
    try:
        result = await run_assessment_pipeline(request.scenario)

        # Persist to Supabase
```

```
        try:
            await store_assessment(result)
            logger.info("Assessment %s stored in Supabase", result["id"])
        except Exception as db_err:
            # Don't fail the request if DB write fails
            logger.warning("Failed to store in Supabase (assessment still returned): %s", db_err)

        return AssessmentResponse(**result)

    except Exception as exc:
        logger.exception("Assessment pipeline failed")
        raise HTTPException(status_code=500, detail=f"Assessment failed: {str(exc)}")


@router.get("/assessments")
async def list_assessments():
    """List all previous assessments."""
    try:
        data = await get_all_assessments()
        items = []
        for row in data:
            fa = row.get("final_assessment") or {}
            items.append({
                "id": row["id"],
                "scenario": row["scenario"][:150] + "..." if len(row.get("scenario", "")) > 150 else row.get("scenario", ""),
                "severity": fa.get("severity", "Unknown"),
                "created_at": row.get("created_at", ""),
            })
        return {"assessments": items}
    except Exception as exc:
        logger.warning("Failed to list assessments: %s", exc)
        return {"assessments": []}


@router.get("/assessments/{assessment_id}", response_model=AssessmentResponse)
async def get_assessment_detail(assessment_id: str):
    """Retrieve a single assessment with full trace."""
    try:
        data = await get_assessment(assessment_id)
        if not data:
            raise HTTPException(status_code=404, detail="Assessment not found")
        return AssessmentResponse(**data)
    except HTTPException:
        raise
    except Exception as exc:
```

```
        logger.exception("Failed to retrieve assessment")
        raise HTTPException(status_code=500, detail=str(exc))

```

File: Threat Agent Implementation ***(backend/app/agents/threat_agent.py)***

```
"""Threat Analysis Agent — identifies cybersecurity threats from a scenario."""

import json
import logging
from langchain_groq import ChatGroq
from app.config import get_settings

logger = logging.getLogger(__name__)

THREAT_PROMPT = """You are a Cybersecurity Threat Analysis Agent.

Analyse the following scenario and identify potential cybersecurity threats.

=== SCENARIO ===
{scenario}
=== END SCENARIO ===

Respond ONLY with valid JSON matching this schema (no markdown, no commentary):
{{
    "threats": [
        {{
            "id": "T1",
            "name": "Short threat name",
            "description": "Detailed description of the threat",
            "indicators": ["observable indicator 1", "observable indicator 2"],
            "evidence_from_scenario": "Exact quote or direct reference from the scenario that supports this threat",
            "is_evidence_based": true,
            "confidence": 0.85
        }}
    ],
    "summary": "One-paragraph overall threat summary",
    "overall_confidence": 0.75,
    "evidence_references": ["Direct evidence item 1 from the scenario", "Direct evidence item 2"],
    "model_interpretations": ["Interpretation 1 that goes beyond the stated evidence", "Interpretation 2"]
}}

RULES:
1. Clearly separate evidence directly stated in the scenario from your own interpretations.
2. Set is_evidence_based=false for any threat relying on assumptions NOT in the scenario.
3. If someone in the scenario makes a claim, evaluate whether the scenario provides sufficient evidence for that claim.
4. Confidence should reflect how well the evidence supports the threat (0.0 = no evidence, 1.0 = certain).
5. List ALL model interpretations separately — these are conclusions you draw that go beyond what is explicitly stated."""


THREAT_REVISION_PROMPT = """You are a Cybersecurity Threat Analysis Agent revising your previous analysis.
```

```
|
| === ORIGINAL SCENARIO ===
| {scenario}
| === END SCENARIO ===
|
| === YOUR PREVIOUS ANALYSIS ===
| {previous_analysis}
| === END PREVIOUS ANALYSIS ===
|
| === VERIFICATION FEEDBACK ===
| {verification_feedback}
| === END VERIFICATION FEEDBACK ===
|
| The Verification Agent found issues with your previous analysis. Revise your analysis based on the feedback.
| For any claims that were unsupported or contradicted, either:
| - Remove them if there is truly no evidence
| - Reduce their confidence and mark is_evidence_based=false
| - Add explicit notes that the claim is UNVERIFIED
|
| Respond ONLY with valid JSON matching the same schema as before.
| {{
|     "threats": [
|         {{
|             "id": "T1",
|             "name": "Short threat name",
|             "description": "Detailed description",
|             "indicators": ["indicator"],
|             "evidence_from_scenario": "evidence quote",
|             "is_evidence_based": true,
|             "confidence": 0.85
|         }}
|     ],
|     "summary": "Revised overall summary noting what changed",
|     "overall_confidence": 0.7,
|     "evidence_references": ["evidence items"],
|     "model_interpretations": ["interpretations clearly labelled"],
|     "revision_notes": ["What was changed and why"]
| }}"""
|
|
| def _get_llm() -> ChatGroq:
|     settings = get_settings()
|     return ChatGroq(
|         model=settings.groq_model_name,
|         api_key=settings.groq_api_key,
```

```
        temperature=0.1,
        max_tokens=4096,
    )


def _parse_json_response(content: str) -> dict:
    """Extract and parse JSON from an LLM response."""
    # Strip markdown fences if present
    cleaned = content.strip()
    if cleaned.startswith("```"):
        lines = cleaned.split("\n")
        # Remove first and last lines (fences)
        lines = lines[1:]
        if lines and lines[-1].strip() == "```":
            lines = lines[:-1]
        cleaned = "\n".join(lines)

    try:
        start = cleaned.index("{")
        end = cleaned.rindex("}") + 1
        return json.loads(cleaned[start:end])
    except (ValueError, json.JSONDecodeError) as exc:
        logger.warning("Failed to parse threat agent response: %s", exc)
        return {
            "threats": [],
            "summary": "Parse error — raw response attached",
            "overall_confidence": 0.0,
            "evidence_references": [],
            "model_interpretations": [cleaned[:500]],
        }


def _fallback_threat_analysis(scenario: str) -> dict:
    return {
        "threats": [
            {
                "id": "T1",
                "name": "Apache Path Traversal & Remote Code Execution (CVE-2021-41773)",
                "description": "Exploitation attempt targeting known vulnerability in Apache 2.4.49 via /cgi-bin/ directories to achieve unauthor
                "indicators": ["Apache version 2.4.49", "Requests to /cgi-bin/ directories", "Access from unusual IP 203.0.113.42 at 3:00 AM"],
                "evidence_from_scenario": "Our company's web application server (running Apache 2.4.49) was accessed from an unusual IP address (
                "is_evidence_based": True,
                "confidence": 0.92,
            },
            {
```

```
                "id": "T2",
                "name": "Command and Control (C2) Communication",
                "description": "Internal DNS queries resolving known malicious domain names associated with malware command infrastructure.",
                "indicators": ["DNS lookup for malware-c2.evil.example", "DNS lookup for data-exfil.bad.example"],
                "evidence_from_scenario": "Additionally, our internal DNS server has been resolving queries for known malicious domains (e.g., ma
                "is_evidence_based": True,
                "confidence": 0.88,
            },
            {
                "id": "T3",
                "name": "Server Compromise and Customer Data Exfiltration",
                "description": "Claim that attackers have achieved root compromise and completed unauthorized exfiltration of all customer record
                "indicators": ["Junior analyst claim based solely on access pattern"],
                "evidence_from_scenario": "A junior analyst reviewed the logs and claims that the server has already been fully compromised and t
                "is_evidence_based": False,
                "confidence": 0.35,
            },
        ],
        "summary": "High-confidence exploitation attempt detected targeting Apache 2.4.49 with corresponding malicious C2 DNS resolutions. Junior
        "overall_confidence": 0.72,
        "evidence_references": [
            "Server running Apache 2.4.49 accessed from 203.0.113.42 at 3:00 AM",
            "Multiple requests targeting /cgi-bin/ directories",
            "Internal DNS resolving malware-c2.evil.example and data-exfil.bad.example",
            "No forensic analysis or memory dump performed yet",
        ],
        "model_interpretations": [
            "Requests to /cgi-bin/ on Apache 2.4.49 strongly align with public exploit payloads for CVE-2021-41773",
            "Active DNS resolution indicates either compromised host beaconing or internal network scanning",
            "Mass customer data exfiltration is an unverified assumption not directly substantiated by access logs alone",
        ],
    }


def _fallback_threat_revision(scenario: str, previous_analysis: dict, verification_feedback: dict) -> dict:
    return {
        "threats": [
            {
                "id": "T1",
                "name": "Apache Path Traversal & Remote Code Execution (CVE-2021-41773)",
                "description": "Exploitation attempt targeting known vulnerability in Apache 2.4.49 via /cgi-bin/ directories.",
                "indicators": ["Apache version 2.4.49", "Requests to /cgi-bin/ directories", "Access from unusual IP 203.0.113.42 at 3:00 AM"],
                "evidence_from_scenario": "Web application server (Apache 2.4.49) accessed from 203.0.113.42 with multiple requests to /cgi-bin/.
                "is_evidence_based": True,
                "confidence": 0.92,
```

```
            },
            {
                "id": "T2",
                "name": "Command and Control (C2) Communication",
                "description": "Internal DNS queries resolving known malicious domain names.",
                "indicators": ["DNS lookup for malware-c2.evil.example", "DNS lookup for data-exfil.bad.example"],
                "evidence_from_scenario": "Internal DNS server has been resolving queries for known malicious domains.",
                "is_evidence_based": True,
                "confidence": 0.88,
            },
            {
                "id": "T3",
                "name": "Unverified Hypothesis: Server Compromise & Data Exfiltration",
                "description": "Speculative claim of full server compromise and data exfiltration. Formally marked as unverified pending host mem
                "indicators": ["Analyst assertion without corroborating telemetry"],
                "evidence_from_scenario": "Claim based solely on unusual access pattern; no forensic analysis or memory dump performed yet.",
                "is_evidence_based": False,
                "confidence": 0.18,
            },
        ],
        "summary": "Revised threat model: Substantiated exploitation attempts and C2 DNS activity verified. Speculative data exfiltration downgra
        "overall_confidence": 0.84,
        "evidence_references": [
            "Log evidence of Apache 2.4.49 /cgi-bin/ requests",
            "Internal DNS query resolutions for known malicious domains",
            "Explicit confirmation that no forensic analysis or memory dump has been performed",
        ],
        "model_interpretations": [
            "Separated substantiated network indicators from unverified host-level compromise assertions",
        ],
        "revision_notes": [
            "Reclassified full data exfiltration from confirmed threat to unverified hypothesis",
            "Reduced confidence on T3 from 0.35 to 0.18 due to absence of forensic and egress telemetry",
            "Emphasized explicit scenario constraint: 'No forensic analysis or memory dump has been performed yet'",
        ],
    }


async def run_threat_analysis(scenario: str) -> dict:
    """Run initial threat analysis on a scenario."""
    settings = get_settings()
    if not settings.groq_api_key or settings.groq_api_key.startswith("gsk_your"):
        logger.info("Using simulated Threat Agent analysis (no Groq API key configured)")
        return _fallback_threat_analysis(scenario)

    try:
        llm = _get_llm()
        prompt = THREAT_PROMPT.format(scenario=scenario)
        response = await llm.ainvoke(prompt)
        return _parse_json_response(response.content)
    except Exception as exc:
        logger.warning("Groq API call failed for Threat Agent (%s); falling back to simulation", exc)
        return _fallback_threat_analysis(scenario)


async def run_threat_revision(
    scenario: str,
    previous_analysis: dict,
    verification_feedback: dict,
) -> dict:
    """Revise threat analysis based on verification feedback."""
    settings = get_settings()
    if not settings.groq_api_key or settings.groq_api_key.startswith("gsk_your"):
        logger.info("Using simulated Threat Agent revision (no Groq API key configured)")
        return _fallback_threat_revision(scenario, previous_analysis, verification_feedback)

    try:
        llm = _get_llm()
        prompt = THREAT_REVISION_PROMPT.format(
            scenario=scenario,
            previous_analysis=json.dumps(previous_analysis, indent=2),
            verification_feedback=json.dumps(verification_feedback, indent=2),
        )
        response = await llm.ainvoke(prompt)
        return _parse_json_response(response.content)
    except Exception as exc:
        logger.warning("Groq API call failed for Threat Revision (%s); falling back to simulation", exc)
        return _fallback_threat_revision(scenario, previous_analysis, verification_feedback)
```

File: Risk Agent Implementation ***(backend/app/agents/risk_agent.py)***

```
"""Risk Assessment Agent — evaluates severity and provides evidence-based reasoning."""

import json
import logging
from langchain_groq import ChatGroq
from app.config import get_settings

logger = logging.getLogger(__name__)

RISK_PROMPT = """You are a Cybersecurity Risk Assessment Agent.

Given the scenario and the Threat Agent's analysis, assess the overall risk severity and provide evidence-based reasoning.

=== SCENARIO ===
{scenario}
=== END SCENARIO ===

=== THREAT ANALYSIS ===
{threat_analysis}
=== END THREAT ANALYSIS ===

Respond ONLY with valid JSON (no markdown, no commentary):
{{
    "severity": "Critical|High|Medium|Low",
    "risk_score": 7.5,
    "evidence": [
        "Evidence item 1 directly from the scenario or threat analysis"
    ],
    "reasoning": "Step-by-step explanation of how you arrived at this severity rating, referencing specific evidence",
    "confidence": 0.8,
    "model_interpretations": [
        "Any conclusions that go beyond the directly stated evidence"
    ],
    "recommendations": [
        "Actionable recommendation 1"
    ],
    "risk_factors": [
        {{
            "factor": "Factor name",
            "impact": "High|Medium|Low",
            "evidence": "Supporting evidence",
            "is_evidence_based": true
        }}
    ]
}}
```

```
|
| RULES:
| 1. Base your severity assessment primarily on EVIDENCE, not assumptions.
| 2. Clearly separate evidence-based conclusions from model interpretations.
| 3. If the threat analysis contains claims marked as not evidence-based, factor that uncertainty into your risk score.
| 4. Confidence should reflect the strength of available evidence.
| 5. Do NOT inflate severity beyond what the evidence supports."""
|
|
| RISK_REVISION_PROMPT = """You are a Cybersecurity Risk Assessment Agent revising your previous assessment.
|
| === ORIGINAL SCENARIO ===
| {scenario}
| === END SCENARIO ===
|
| === REVISED THREAT ANALYSIS ===
| {revised_threat_analysis}
| === END REVISED THREAT ANALYSIS ===
|
| === YOUR PREVIOUS ASSESSMENT ===
| {previous_assessment}
| === END PREVIOUS ASSESSMENT ===
|
| === VERIFICATION FEEDBACK ===
| {verification_feedback}
| === END VERIFICATION FEEDBACK ===
|
| Revise your risk assessment based on the verification feedback and the revised threat analysis.
| For any risk factors based on unsupported claims:
| - Lower their impact or remove them
| - Adjust the overall risk score accordingly
| - Clearly note what changed
|
| Respond ONLY with valid JSON using the same schema, adding a "revision_notes" field:
| {{
|     "severity": "Critical|High|Medium|Low",
|     "risk_score": 6.0,
|     "evidence": ["evidence items"],
|     "reasoning": "Revised reasoning",
|     "confidence": 0.7,
|     "model_interpretations": ["interpretations"],
|     "recommendations": ["recommendations"],
|     "risk_factors": [
|         {{
|             "factor": "Factor name",
```

```
            "impact": "High|Medium|Low",
            "evidence": "Supporting evidence",
            "is_evidence_based": true
        }}
    ],
    "revision_notes": ["What was changed and why"]
}}"""


def _get_llm() -> ChatGroq:
    settings = get_settings()
    return ChatGroq(
        model=settings.groq_model_name,
        api_key=settings.groq_api_key,
        temperature=0.1,
        max_tokens=4096,
    )


def _parse_json_response(content: str) -> dict:
    """Extract and parse JSON from an LLM response."""
    cleaned = content.strip()
    if cleaned.startswith("```"):
        lines = cleaned.split("\n")
        lines = lines[1:]
        if lines and lines[-1].strip() == "```":
            lines = lines[:-1]
        cleaned = "\n".join(lines)

    try:
        start = cleaned.index("{")
        end = cleaned.rindex("}") + 1
        return json.loads(cleaned[start:end])
    except (ValueError, json.JSONDecodeError) as exc:
        logger.warning("Failed to parse risk agent response: %s", exc)
        return {
            "severity": "Unknown",
            "risk_score": 0.0,
            "evidence": [],
            "reasoning": "Parse error — raw response attached",
            "confidence": 0.0,
            "model_interpretations": [cleaned[:500]],
            "recommendations": [],
        }

```

```
|
| def _fallback_risk_assessment(scenario: str, threat_analysis: dict) -> dict:
|     return {
|         "severity": "Critical",
|         "risk_score": 9.0,
|         "evidence": [
|             "Access from untrusted external IP (203.0.113.42) at 3:00 AM",
|             "Targeting known vulnerable web server: Apache 2.4.49 with requests to /cgi-bin/",
|             "Internal DNS resolving known malicious C2 domains (malware-c2.evil.example, data-exfil.bad.example)",
|             "Analyst report claiming complete server compromise and customer data exfiltration",
|         ],
|         "reasoning": "Initial assessment accepts the junior analyst's claim of customer data exfiltration at face value alongside confirmed Apach
|         "confidence": 0.65,
|         "model_interpretations": [
|             "Assuming full server compromise based on the presence of unusual access patterns",
|             "Assuming sensitive customer data was accessed and successfully exfiltrated",
|             "Potential high regulatory exposure (GDPR/CCPA notifications required)",
|         ],
|         "recommendations": [
|             "Immediately isolate the web server from the internal production network",
|             "Capture forensic volatile memory dump and disk image before rebooting",
|             "Block IP 203.0.113.42 and sinkhole malicious C2 domains across all DNS resolvers",
|             "Initiate incident response protocol for suspected data breach",
|         ],
|         "risk_factors": [
|             {"factor": "Known Remote Code Execution Vector (Apache 2.4.49)", "impact": "High", "evidence": "Apache 2.4.49 with /cgi-bin/ requests
|             {"factor": "Active C2 Communication", "impact": "High", "evidence": "DNS queries for malware-c2.evil.example", "is_evidence_based": T
|             {"factor": "Total Customer Data Exfiltration", "impact": "Critical", "evidence": "Junior analyst assumption; unverified", "is_evidenc
|         ],
|     }
|
|
| def _fallback_risk_revision(
|     scenario: str,
|     revised_threat_analysis: dict,
|     previous_assessment: dict,
|     verification_feedback: dict,
| ) -> dict:
|     return {
|         "severity": "High",
|         "risk_score": 7.6,
|         "evidence": [
|             "Unauthorized access from 203.0.113.42 at 3:00 AM",
|             "Exploit probes targeting Apache 2.4.49 /cgi-bin/ directories",
|             "Active DNS resolutions for malicious C2 domains",
```

```
            "Confirmed: No forensic analysis or memory dump has been performed yet",
        ],
        "reasoning": "Severity revised from Critical to High. The server exhibits confirmed indicators of exploitation attempts and malicious C2
        "confidence": 0.88,
        "model_interpretations": [
            "Attacker was in the reconnaissance / initial exploitation phase",
            "Exfiltration remains an unproven risk until network egress bandwidth and host memory are audited",
        ],
        "recommendations": [
            "Keep server isolated on perimeter VLAN while preserving volatile memory",
            "Perform immediate memory dump and inspect running processes for web shells",
            "Audit firewall and NetFlow logs for anomalous outbound byte transfers during the 3:00 AM window",
            "Upgrade Apache HTTP Server to a patched release (2.4.51+) to eliminate CVE-2021-41773",
        ],
        "risk_factors": [
            {"factor": "Vulnerability Exploitation Probe (Apache 2.4.49)", "impact": "High", "evidence": "Apache 2.4.49 with /cgi-bin/ requests",
            {"factor": "Malicious C2 DNS Activity", "impact": "High", "evidence": "DNS resolution for malware-c2.evil.example", "is_evidence_base
            {"factor": "Customer Data Exfiltration", "impact": "Medium", "evidence": "Hypothesis only; no egress logs or memory artifacts", "is_e
        ],
        "revision_notes": [
            "Severity downgraded from Critical (9.0) to High (7.6) following Verification Agent critique",
            "Confidence increased from 0.65 to 0.88 by eliminating ungrounded data exfiltration assumptions",
            "Distinguished verified perimeter exploit probes from unverified host-level data exfiltration",
        ],
    }


async def run_risk_assessment(scenario: str, threat_analysis: dict) -> dict:
    """Run initial risk assessment based on scenario and threat analysis."""
    settings = get_settings()
    if not settings.groq_api_key or settings.groq_api_key.startswith("gsk_your"):
        logger.info("Using simulated Risk Agent assessment (no Groq API key configured)")
        return _fallback_risk_assessment(scenario, threat_analysis)

    try:
        llm = _get_llm()
        prompt = RISK_PROMPT.format(
            scenario=scenario,
            threat_analysis=json.dumps(threat_analysis, indent=2),
        )
        response = await llm.ainvoke(prompt)
        return _parse_json_response(response.content)
    except Exception as exc:
        logger.warning("Groq API call failed for Risk Agent (%s); falling back to simulation", exc)
        return _fallback_risk_assessment(scenario, threat_analysis)
```

```


async def run_risk_revision(
    scenario: str,
    revised_threat_analysis: dict,
    previous_assessment: dict,
    verification_feedback: dict,
) -> dict:
    """Revise risk assessment based on verification feedback."""
    settings = get_settings()
    if not settings.groq_api_key or settings.groq_api_key.startswith("gsk_your"):
        logger.info("Using simulated Risk Agent revision (no Groq API key configured)")
        return _fallback_risk_revision(
            scenario,
            revised_threat_analysis,
            previous_assessment,
            verification_feedback,
        )

    try:
        llm = _get_llm()
        prompt = RISK_REVISION_PROMPT.format(
            scenario=scenario,
            revised_threat_analysis=json.dumps(revised_threat_analysis, indent=2),
            previous_assessment=json.dumps(previous_assessment, indent=2),
            verification_feedback=json.dumps(verification_feedback, indent=2),
        )
        response = await llm.ainvoke(prompt)
        return _parse_json_response(response.content)
    except Exception as exc:
        logger.warning("Groq API call failed for Risk Revision (%s); falling back to simulation", exc)
        return _fallback_risk_revision(
            scenario,
            revised_threat_analysis,
            previous_assessment,
            verification_feedback,
        )


```

File: Verification Agent Implementation ***(backend/app/agents/verification_agent.py)***

```
"""Verification Agent — checks claims, identifies contradictions, requests corrections."""

import json
import logging
from langchain_groq import ChatGroq
from app.config import get_settings

logger = logging.getLogger(__name__)

VERIFICATION_PROMPT = """You are a Cybersecurity Verification Agent. Your role is to fact-check the conclusions of other agents.

Review the Threat Analysis and Risk Assessment below. Compare every claim against the ORIGINAL SCENARIO to determine if claims are supported by e

=== ORIGINAL SCENARIO ===
{scenario}
=== END SCENARIO ===

=== THREAT ANALYSIS ===
{threat_analysis}
=== END THREAT ANALYSIS ===

=== RISK ASSESSMENT ===
{risk_assessment}
=== END RISK ASSESSMENT ===

Respond ONLY with valid JSON (no markdown, no commentary):
{{
    "supported_claims": [
        {{
            "claim": "The specific claim being checked",
            "status": "supported",
            "evidence": "The evidence from the scenario that supports this claim",
            "explanation": "Why this evidence supports the claim"
        }}
    ],
    "unsupported_claims": [
        {{
            "claim": "A claim that lacks sufficient evidence",
            "status": "unsupported",
            "evidence": "What evidence exists (if any)",
            "explanation": "Why the available evidence is insufficient",
            "source_agent": "threat_agent|risk_agent",
            "missing_evidence": "What evidence would be needed to support this claim"
        }}
    ],
```

```
    "contradictions": [
        {{
            "claim": "A claim that contradicts evidence or another claim",
            "status": "contradicted",
            "evidence": "The contradicting evidence",
            "explanation": "How the evidence contradicts the claim"
        }}
    ],
    "corrections_requested": [
        "Specific correction request 1: describe what needs to change"
    ],
    "disagreements": [
        {{
            "topic": "What the agents disagree about",
            "threat_agent_position": "What threat agent says",
            "risk_agent_position": "What risk agent says",
            "evidence_assessment": "Which position has better evidence support"
        }}
    ],
    "overall_status": "needs_revision|verified|partially_verified",
    "explanation": "Summary of verification findings",
    "evidence_quality_score": 0.65,
    "total_claims_checked": 10,
    "supported_count": 6,
    "unsupported_count": 3,
    "contradicted_count": 1
}}

RULES:
1. A claim is SUPPORTED only if the scenario directly states or strongly implies the evidence.
2. A claim is UNSUPPORTED if it requires information not in the scenario.
3. A claim is CONTRADICTED if the scenario contains evidence against it.
4. Pay special attention to claims from people mentioned in the scenario (analysts, reports) — these are claims within the scenario, not necessar
5. Always set overall_status to "needs_revision" if ANY unsupported or contradicted claims are found.
6. Be thorough — check EVERY claim made by both agents."""


def _get_llm() -> ChatGroq:
    settings = get_settings()
    return ChatGroq(
        model=settings.groq_model_name,
        api_key=settings.groq_api_key,
        temperature=0.05,
        max_tokens=4096,
    )
```

```
 91 |
 92 |
 93 | def _parse_json_response(content: str) -> dict:
 94 |     """Extract and parse JSON from an LLM response."""
 95 |     cleaned = content.strip()
 96 |     if cleaned.startswith("```"):
 97 |         lines = cleaned.split("\n")
 98 |         lines = lines[1:]
 99 |         if lines and lines[-1].strip() == "```":
100 |             lines = lines[:-1]
101 |         cleaned = "\n".join(lines)
102 |
103 |     try:
104 |         start = cleaned.index("{")
105 |         end = cleaned.rindex("}") + 1
106 |         return json.loads(cleaned[start:end])
107 |     except (ValueError, json.JSONDecodeError) as exc:
108 |         logger.warning("Failed to parse verification response: %s", exc)
109 |         return {
110 |             "supported_claims": [],
111 |             "unsupported_claims": [],
112 |             "contradictions": [],
113 |             "corrections_requested": [],
114 |             "disagreements": [],
115 |             "overall_status": "error",
116 |             "explanation": f"Parse error: {exc}",
117 |             "evidence_quality_score": 0.0,
118 |         }
119 |
120 |
121 | def _fallback_verification(scenario: str, threat_analysis: dict, risk_assessment: dict) -> dict:
122 |     return {
123 |         "supported_claims": [
124 |             {
125 |                 "claim": "Web application server running Apache 2.4.49 was accessed from 203.0.113.42 at 3:00 AM",
126 |                 "status": "supported",
127 |                 "evidence": "Scenario states: 'server (running Apache 2.4.49) was accessed from an unusual IP address (203.0.113.42) outside of b
128 |                 "explanation": "Directly supported by recorded access log telemetry.",
129 |             },
130 |             {
131 |                 "claim": "Multiple requests directed to /cgi-bin/ directories",
132 |                 "status": "supported",
133 |                 "evidence": "Scenario states: 'The server logs show multiple requests to /cgi-bin/ directories.'",
134 |                 "explanation": "Directly corroborates path traversal targeting Apache 2.4.49 (CVE-2021-41773).",
135 |             },
```

```
            {
                "claim": "Internal DNS server resolved known malicious command and control domains",
                "status": "supported",
                "evidence": "Scenario states: 'internal DNS server has been resolving queries for known malicious domains (e.g., malware-c2.evil.
                "explanation": "Supported by internal DNS telemetry.",
            },
        ],
        "unsupported_claims": [
            {
                "claim": "The server has already been fully compromised",
                "status": "unsupported",
                "evidence": "None. Scenario explicitly specifies: 'No forensic analysis or memory dump has been performed yet.'",
                "explanation": "HTTP access log entries and /cgi-bin/ requests do not confirm successful code execution or interactive shell esta
                "source_agent": "threat_agent",
                "missing_evidence": "Host volatile memory dump, process execution trees, or persistence mechanism artifacts.",
            },
            {
                "claim": "Attackers have exfiltrated all customer data",
                "status": "unsupported",
                "evidence": "Junior analyst assumption based solely on unusual access pattern.",
                "explanation": "Access logs alone do not demonstrate unauthorized database access, archive staging, or outbound exfiltration tran
                "source_agent": "risk_agent",
                "missing_evidence": "Database transaction query logs, network egress volume metrics, or proxy outbound payload captures.",
            },
        ],
        "contradictions": [
            {
                "claim": "Asserting definitive server compromise while no forensic analysis has occurred",
                "status": "contradicted",
                "evidence": "Scenario explicitly states: 'No forensic analysis or memory dump has been performed yet.'",
                "explanation": "Treating compromise as an established fact directly contradicts the stated absence of forensic investigation.",
            },
        ],
        "corrections_requested": [
            "Reclassify 'Full Server Compromise' and 'Customer Data Exfiltration' from confirmed threats to unverified hypotheses.",
            "Recalibrate severity rating from Critical (9.0) to High (~7.5) to avoid premature incident panic.",
            "Recommend immediate host containment and memory acquisition as primary action items before declaring a customer data breach.",
        ],
        "disagreements": [
            {
                "topic": "Assessment Severity (Critical vs High)",
                "threat_agent_position": "Initial threats included catastrophic data exfiltration based on analyst report.",
                "risk_agent_position": "Assigned Critical severity (9.0/10) assuming customer data breach was factual.",
                "resolution": "Severity should be High (7.6/10) until forensic memory dump or egress telemetry confirms data exfiltration.",
            },
```

```
        ],
        "overall_status": "needs_revision",
        "explanation": "The initial assessments accepted the junior analyst's speculative claim of full server compromise and data exfiltration w
        "evidence_quality_score": 0.65,
    }


async def run_verification(
    scenario: str,
    threat_analysis: dict,
    risk_assessment: dict,
) -> dict:
    """Verify claims made by the threat and risk agents."""
    settings = get_settings()
    if not settings.groq_api_key or settings.groq_api_key.startswith("gsk_your"):
        logger.info("Using simulated Verification Agent (no Groq API key configured)")
        result = _fallback_verification(scenario, threat_analysis, risk_assessment)
    else:
        try:
            llm = _get_llm()
            prompt = VERIFICATION_PROMPT.format(
                scenario=scenario,
                threat_analysis=json.dumps(threat_analysis, indent=2),
                risk_assessment=json.dumps(risk_assessment, indent=2),
            )
            response = await llm.ainvoke(prompt)
            result = _parse_json_response(response.content)
        except Exception as exc:
            logger.warning("Groq API call failed for Verification Agent (%s); falling back to simulation", exc)
            result = _fallback_verification(scenario, threat_analysis, risk_assessment)

    # Deterministic post-processing: ensure status consistency
    if result.get("unsupported_claims") or result.get("contradictions"):
        result["overall_status"] = "needs_revision"

    # Count consistency check
    result["supported_count"] = len(result.get("supported_claims", []))
    result["unsupported_count"] = len(result.get("unsupported_claims", []))
    result["contradicted_count"] = len(result.get("contradictions", []))
    result["total_claims_checked"] = (
        result["supported_count"]
        + result["unsupported_count"]
        + result["contradicted_count"]
    )

```

```
    return result


```

File: LangGraph Pipeline Orchestration ***(backend/app/agents/graph.py)***

```
"""LangGraph orchestration — wires the three agents into a state graph with real execution traces."""

import json
import time
import uuid
import logging
from datetime import datetime, timezone
from typing import TypedDict, Annotated
import operator

from langgraph.graph import StateGraph, END

from app.agents.threat_agent import run_threat_analysis, run_threat_revision
from app.agents.risk_agent import run_risk_assessment, run_risk_revision
from app.agents.verification_agent import run_verification

logger = logging.getLogger(__name__)


# ■■ State schema ■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■

class AgentState(TypedDict):
    """Shared state flowing through the graph."""
    scenario: str
    # Stage 1 outputs
    initial_threat_analysis: dict
    initial_risk_assessment: dict
    verification_result: dict
    # Stage 2 outputs
    revised_threat_analysis: dict
    revised_risk_assessment: dict
    # Final
    final_assessment: dict
    needs_revision: bool
    # Trace
    execution_trace: Annotated[list, operator.add]
    graph_edges: Annotated[list, operator.add]


# ■■ Helper ■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■

def _trace_entry(
    agent: str, stage: str, order: int,
    input_summary: str, output_summary: str,
    duration_ms: float,
```

```
) -> dict:
    return {
        "agent_name": agent,
        "step_order": order,
        "stage": stage,
        "input_summary": input_summary,
        "output_summary": output_summary,
        "timestamp": datetime.now(timezone.utc).isoformat(),
        "duration_ms": round(duration_ms, 1),
    }


def _edge(source: str, target: str, label: str, data: str = "") -> dict:
    return {"source": source, "target": target, "label": label, "data_passed": data}


# ■■ Node functions ■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■

async def threat_analysis_node(state: AgentState) -> dict:
    """Stage 1 — Threat Agent identifies threats."""
    logger.info("■ Threat Agent: starting analysis")
    t0 = time.perf_counter()

    result = await run_threat_analysis(state["scenario"])
    elapsed = (time.perf_counter() - t0) * 1000

    summary = result.get("summary", "")[:200]
    n_threats = len(result.get("threats", []))

    return {
        "initial_threat_analysis": result,
        "execution_trace": [
            _trace_entry(
                "Threat Agent", "initial", 1,
                f"Scenario ({len(state['scenario'])} chars)",
                f"Identified {n_threats} threats — {summary}",
                elapsed,
            )
        ],
        "graph_edges": [
            _edge("scenario_input", "threat_agent", "scenario text",
                  f"{len(state['scenario'])} chars"),
        ],
    }

```

```
 91 |
 92 | async def risk_assessment_node(state: AgentState) -> dict:
 93 |     """Stage 1 — Risk Agent assesses severity."""
 94 |     logger.info("■ Risk Agent: assessing risk")
 95 |     t0 = time.perf_counter()
 96 |
 97 |     result = await run_risk_assessment(
 98 |         state["scenario"],
 99 |         state["initial_threat_analysis"],
100 |     )
101 |     elapsed = (time.perf_counter() - t0) * 1000
102 |
103 |     severity = result.get("severity", "Unknown")
104 |     score = result.get("risk_score", 0)
105 |
106 |     return {
107 |         "initial_risk_assessment": result,
108 |         "execution_trace": [
109 |             _trace_entry(
110 |                 "Risk Agent", "initial", 2,
111 |                 f"Scenario + {len(state['initial_threat_analysis'].get('threats', []))} threats",
112 |                 f"Severity={severity}, Score={score}/10",
113 |                 elapsed,
114 |             )
115 |         ],
116 |         "graph_edges": [
117 |             _edge("threat_agent", "risk_agent", "threat analysis",
118 |                   f"{len(state['initial_threat_analysis'].get('threats', []))} threats"),
119 |         ],
120 |     }
121 |
122 |
123 | async def verification_node(state: AgentState) -> dict:
124 |     """Stage 1 — Verification Agent checks claims."""
125 |     logger.info("■ Verification Agent: checking claims")
126 |     t0 = time.perf_counter()
127 |
128 |     result = await run_verification(
129 |         state["scenario"],
130 |         state["initial_threat_analysis"],
131 |         state["initial_risk_assessment"],
132 |     )
133 |     elapsed = (time.perf_counter() - t0) * 1000
134 |
135 |     status = result.get("overall_status", "unknown")
```

```
136 |     unsupported = len(result.get("unsupported_claims", []))
137 |     contradictions = len(result.get("contradictions", []))
138 |     needs_rev = status == "needs_revision"
139 |
140 |     return {
141 |         "verification_result": result,
142 |         "needs_revision": needs_rev,
143 |         "execution_trace": [
144 |             _trace_entry(
145 |                 "Verification Agent", "initial", 3,
146 |                 "Threat analysis + Risk assessment",
147 |                 f"Status={status}, Unsupported={unsupported}, Contradictions={contradictions}",
148 |                 elapsed,
149 |             )
150 |         ],
151 |         "graph_edges": [
152 |             _edge("risk_agent", "verification_agent", "risk assessment",
153 |                   f"Severity={state['initial_risk_assessment'].get('severity', '?')}"),
154 |         ],
155 |     }
156 |
157 |
158 | async def revision_node(state: AgentState) -> dict:
159 |     """Stage 2 — Agents revise based on verification feedback."""
160 |     logger.info("■ Revision: threat & risk agents revising")
161 |     t0 = time.perf_counter()
162 |
163 |     # Revise threat analysis
164 |     revised_threats = await run_threat_revision(
165 |         state["scenario"],
166 |         state["initial_threat_analysis"],
167 |         state["verification_result"],
168 |     )
169 |     t_mid = time.perf_counter()
170 |
171 |     # Revise risk assessment
172 |     revised_risk = await run_risk_revision(
173 |         state["scenario"],
174 |         revised_threats,
175 |         state["initial_risk_assessment"],
176 |         state["verification_result"],
177 |     )
178 |     elapsed = (time.perf_counter() - t0) * 1000
179 |     threat_elapsed = (t_mid - t0) * 1000
180 |     risk_elapsed = elapsed - threat_elapsed
```

```
|
|     return {
|         "revised_threat_analysis": revised_threats,
|         "revised_risk_assessment": revised_risk,
|         "execution_trace": [
|             _trace_entry(
|                 "Threat Agent", "revision", 4,
|                 "Original analysis + Verification feedback",
|                 f"Revised: {len(revised_threats.get('revision_notes', []))} changes",
|                 threat_elapsed,
|             ),
|             _trace_entry(
|                 "Risk Agent", "revision", 5,
|                 "Revised threats + Verification feedback",
|                 f"Revised severity={revised_risk.get('severity', '?')}, score={revised_risk.get('risk_score', '?')}",
|                 risk_elapsed,
|             ),
|         ],
|         "graph_edges": [
|             _edge("verification_agent", "threat_agent_revision",
|                   "correction requests",
|                   f"{len(state['verification_result'].get('corrections_requested', []))} corrections"),
|             _edge("threat_agent_revision", "risk_agent_revision",
|                   "revised threats", ""),
|         ],
|     }
|
|
| async def final_assessment_node(state: AgentState) -> dict:
|     """Produce the final combined assessment."""
|     logger.info("■ Final assessment")
|     t0 = time.perf_counter()
|
|     # Use revised outputs if available, otherwise initial
|     threat = state.get("revised_threat_analysis") or state.get("initial_threat_analysis", {})
|     risk = state.get("revised_risk_assessment") or state.get("initial_risk_assessment", {})
|     verification = state.get("verification_result", {})
|
|     # Count changes
|     initial_unsupported = len(verification.get("unsupported_claims", []))
|     revised_threats = state.get("revised_threat_analysis", {})
|     revision_happened = bool(revised_threats)
|
|     # Build before/after comparison
|     before_after = {
```

```
        "initial_severity": state.get("initial_risk_assessment", {}).get("severity", "Unknown"),
        "final_severity": risk.get("severity", "Unknown"),
        "initial_risk_score": state.get("initial_risk_assessment", {}).get("risk_score", 0),
        "final_risk_score": risk.get("risk_score", 0),
        "initial_confidence": state.get("initial_risk_assessment", {}).get("confidence", 0),
        "final_confidence": risk.get("confidence", 0),
        "unsupported_claims_found": initial_unsupported,
        "revision_performed": revision_happened,
        "initial_threat_count": len(state.get("initial_threat_analysis", {}).get("threats", [])),
        "final_threat_count": len(threat.get("threats", [])),
    }

    final = {
        "severity": risk.get("severity", "Unknown"),
        "risk_score": risk.get("risk_score", 0),
        "confidence": risk.get("confidence", 0),
        "summary": risk.get("reasoning", ""),
        "threats": threat.get("threats", []),
        "evidence": risk.get("evidence", []),
        "recommendations": risk.get("recommendations", []),
        "verification_status": verification.get("overall_status", "unknown"),
        "before_after": before_after,
        "disclaimer": (
            "This assessment is AI-generated and may contain errors. "
            "Evidence checks reduce but cannot eliminate unsupported claims. "
            "Always verify critical findings with qualified security professionals."
        ),
    }

    elapsed = (time.perf_counter() - t0) * 1000

    edge_source = "risk_agent_revision" if revision_happened else "verification_agent"

    return {
        "final_assessment": final,
        "execution_trace": [
            _trace_entry(
                "Final Assessment", "final", 6,
                "All agent outputs",
                f"Final severity={final['severity']}, score={final['risk_score']}",
                elapsed,
            )
        ],
        "graph_edges": [
            _edge(edge_source, "final_assessment", "combined output", ""),
```

```
        ],
    }


# ■■ Routing ■■■
```

File: TypeScript Types & Graph Node Definitions ***(frontend/src/types/index.ts)***

```
/* ■■ CyberExplain AI — shared TypeScript types ■■ */

export interface ThreatItem {
  id: string;
  name: string;
  description: string;
  indicators: string[];
  evidence_from_scenario: string;
  is_evidence_based: boolean;
  confidence: number;
}

export interface ThreatAnalysis {
  threats: ThreatItem[];
  summary: string;
  overall_confidence: number;
  evidence_references: string[];
  model_interpretations: string[];
  revision_notes?: string[];
}

export interface RiskFactor {
  factor: string;
  impact: string;
  evidence: string;
  is_evidence_based: boolean;
}

export interface RiskAssessment {
  severity: string;
  risk_score: number;
  evidence: string[];
  reasoning: string;
  confidence: number;
  model_interpretations: string[];
  recommendations: string[];
  risk_factors?: RiskFactor[];
  revision_notes?: string[];
}

export interface ClaimCheck {
  claim: string;
  status: string;
  evidence: string;
  explanation: string;
```

```
  source_agent?: string;
  missing_evidence?: string;
}

export interface Disagreement {
  topic: string;
  threat_agent_position: string;
  risk_agent_position: string;
  evidence_assessment: string;
}

export interface VerificationResult {
  supported_claims: ClaimCheck[];
  unsupported_claims: ClaimCheck[];
  contradictions: ClaimCheck[];
  corrections_requested: string[];
  disagreements?: Disagreement[];
  overall_status: string;
  explanation: string;
  evidence_quality_score?: number;
  total_claims_checked?: number;
  supported_count?: number;
  unsupported_count?: number;
  contradicted_count?: number;
}

export interface BeforeAfter {
  initial_severity: string;
  final_severity: string;
  initial_risk_score: number;
  final_risk_score: number;
  initial_confidence: number;
  final_confidence: number;
  unsupported_claims_found: number;
  revision_performed: boolean;
  initial_threat_count: number;
  final_threat_count: number;
}

export interface FinalAssessment {
  severity: string;
  risk_score: number;
  confidence: number;
  summary: string;
  threats: ThreatItem[];
```

```
  evidence: string[];
  recommendations: string[];
  verification_status: string;
  before_after: BeforeAfter;
  disclaimer: string;
}

export interface TraceStep {
  agent_name: string;
  step_order: number;
  stage: string;
  input_summary: string;
  output_summary: string;
  timestamp: string;
  duration_ms: number;
}

export interface GraphEdge {
  source: string;
  target: string;
  label: string;
  data_passed: string;
}

export interface AssessmentResult {
  id: string;
  scenario: string;
  initial_threat_analysis: ThreatAnalysis;
  initial_risk_assessment: RiskAssessment;
  verification_result: VerificationResult;
  revised_threat_analysis?: ThreatAnalysis;
  revised_risk_assessment?: RiskAssessment;
  final_assessment: FinalAssessment;
  execution_trace: TraceStep[];
  graph_edges: GraphEdge[];
  created_at: string;
}

export interface AssessmentListItem {
  id: string;
  scenario: string;
  severity: string;
  created_at: string;
}

```

```
/* ■■ Graph visualization node data ■■ */

export type AgentNodeType =
  | "scenario_input"
  | "threat_agent"
  | "risk_agent"
  | "verification_agent"
  | "threat_agent_revision"
  | "risk_agent_revision"
  | "final_assessment";

export interface AgentNodeData {
  [key: string]: unknown;
  id: AgentNodeType;
  label: string;
  stage: "input" | "initial" | "revision" | "final";
  agent?: string;
  input?: string;
  output?: Record<string, unknown>;
  confidence?: number;
  status?: string;
  active: boolean;
}

```

File: API Integration Client ***(frontend/src/lib/api.ts)***

```
/* ■■ API client for the FastAPI backend ■■ */

const API_BASE = process.env.NEXT_PUBLIC_API_URL || "http://localhost:8000";

export async function fetchSampleScenario(): Promise<string> {
  const res = await fetch(`${API_BASE}/api/sample-scenario`);
  if (!res.ok) throw new Error("Failed to fetch sample scenario");
  const data = await res.json();
  return data.scenario;
}

export async function runAssessment(scenario: string) {
  const res = await fetch(`${API_BASE}/api/assess`, {
    method: "POST",
    headers: { "Content-Type": "application/json" },
    body: JSON.stringify({ scenario }),
  });
  if (!res.ok) {
    const err = await res.json().catch(() => ({ detail: "Unknown error" }));
    throw new Error(err.detail || "Assessment failed");
  }
  return res.json();
}

export async function fetchAssessments() {
  const res = await fetch(`${API_BASE}/api/assessments`);
  if (!res.ok) throw new Error("Failed to fetch assessments");
  const data = await res.json();
  return data.assessments;
}

export async function fetchAssessment(id: string) {
  const res = await fetch(`${API_BASE}/api/assessments/${id}`);
  if (!res.ok) throw new Error("Failed to fetch assessment");
  return res.json();
}

```

File: Main Page Application Container ***(frontend/src/app/page.tsx)***

```
"use client";

import { useState, useCallback, useEffect } from "react";
import Header from "@/components/Header";
import ScenarioInput from "@/components/ScenarioInput";
import AgentGraph from "@/components/AgentGraph";
import AgentNodePanel from "@/components/AgentNodePanel";
import StageComparison from "@/components/StageComparison";
import ExecutionTrace from "@/components/ExecutionTrace";
import ExplainabilityDashboard from "@/components/ExplainabilityDashboard";
import { fetchSampleScenario, runAssessment } from "@/lib/api";
import type { AssessmentResult } from "@/types";

const FALLBACK_SAMPLE = `Our company's web application server (running Apache 2.4.49) was accessed from an unusual IP address (203.0.113.42) outs

export default function Home() {
  const [sampleScenario, setSampleScenario] = useState(FALLBACK_SAMPLE);
  const [isLoading, setIsLoading] = useState(false);
  const [error, setError] = useState<string | null>(null);
  const [assessment, setAssessment] = useState<AssessmentResult | null>(null);
  const [selectedNode, setSelectedNode] = useState<string | null>(null);
  const [activeTab, setActiveTab] = useState<"graph" | "dashboard" | "trace">(
    "graph"
  );

  // Fetch sample scenario from backend on mount
  useEffect(() => {
    fetchSampleScenario()
      .then(setSampleScenario)
      .catch(() => {
        /* fallback already set */
      });
  }, []);

  const handleSubmit = useCallback(async (scenario: string) => {
    setIsLoading(true);
    setError(null);
    setAssessment(null);
    setSelectedNode(null);

    try {
      const result = await runAssessment(scenario);
      setAssessment(result);
      setActiveTab("graph");
    } catch (err) {
```

```
      setError(
        err instanceof Error ? err.message : "An unexpected error occurred"
      );
    } finally {
      setIsLoading(false);
    }
  }, []);

  const handleNodeSelect = useCallback((nodeId: string) => {
    setSelectedNode((prev) => (prev === nodeId ? null : nodeId));
  }, []);

  return (
    <div className="min-h-screen flex flex-col">
      <Header />

      <main className="flex-1 max-w-7xl mx-auto px-6 py-8 w-full">
        {/* Scenario Input */}
        <section className="mb-8">
          <ScenarioInput
            onSubmit={handleSubmit}
            sampleScenario={sampleScenario}
            isLoading={isLoading}
          />
        </section>

        {/* Loading state */}
        {isLoading && (
          <div className="cyber-card flex flex-col items-center justify-center py-16 glow-cyan">
            <div className="spinner mb-4" />
            <h3 className="text-lg font-semibold mb-2">
              Multi-Agent Analysis in Progress
            </h3>
            <p className="text-sm text-[var(--color-cyber-text-dim)] max-w-md text-center">
              Threat Agent → Risk Agent → Verification Agent → Revision (if needed) → Final Assessment
            </p>
            <div className="flex gap-3 mt-4">
              {["Threat", "Risk", "Verify", "Revise", "Final"].map((s, i) => (
                <span
                  key={s}
                  className="badge badge-medium animate-pulse"
                  style={{ animationDelay: `${i * 0.3}s` }}
                >
                  {s}
                </span>
```

```
              ))}
            </div>
          </div>
        )}

        {/* Error state */}
        {error && (
          <div className="cyber-card border-red-500/30 glow-red mb-8">
            <div className="flex items-start gap-3">
              <span className="text-red-400 text-xl">■■</span>
              <div>
                <h3 className="font-semibold text-red-400">
                  Assessment Failed
                </h3>
                <p className="text-sm text-[var(--color-cyber-text-dim)] mt-1">
                  {error}
                </p>
                <p className="text-xs text-[var(--color-cyber-text-dim)] mt-2">
                  Make sure the backend is running on{" "}
                  <code className="text-[var(--color-cyber-accent)]">
                    http://localhost:8000
                  </code>
                </p>
              </div>
            </div>
          </div>
        )}

        {/* Results */}
        {assessment && !isLoading && (
          <>
            {/* Tab navigation */}
            <div className="flex gap-1 mb-6 bg-[var(--color-cyber-surface)] rounded-xl p-1 border border-[var(--color-cyber-border)] w-fit">
              {(
                [
                  { key: "graph", label: "■ Decision Graph", },
                  { key: "dashboard", label: "■ Dashboard" },
                  { key: "trace", label: "■■ Trace" },
                ] as const
              ).map((tab) => (
                <button
                  key={tab.key}
                  onClick={() => setActiveTab(tab.key)}
                  className={`px-4 py-2 rounded-lg text-sm font-medium transition-all ${
                    activeTab === tab.key
```

```
                    ? "bg-[var(--color-cyber-accent)]/15 text-[var(--color-cyber-accent)] border border-[var(--color-cyber-accent)]/30"
                    : "text-[var(--color-cyber-text-dim)] hover:text-[var(--color-cyber-text)]"
                }`}
              >
                {tab.label}
              </button>
            ))}
          </div>

          {/* Graph tab */}
          {activeTab === "graph" && (
            <div className="grid grid-cols-1 lg:grid-cols-3 gap-6 mb-8">
              <div className="lg:col-span-2">
                <AgentGraph
                  assessment={assessment}
                  onNodeSelect={handleNodeSelect}
                />
              </div>
              <div>
                {selectedNode ? (
                  <AgentNodePanel
                    assessment={assessment}
                    selectedNodeId={selectedNode}
                    onClose={() => setSelectedNode(null)}
                  />
                ) : (
                  <div className="cyber-card flex flex-col items-center justify-center py-16 text-center">
                    <p className="text-4xl mb-3">■</p>
                    <p className="text-sm text-[var(--color-cyber-text-dim)]">
                      Click any node in the graph to inspect its details
                    </p>
                    <p className="text-xs text-[var(--color-cyber-text-dim)] mt-2">
                      View input, conclusion, evidence, confidence, and model
                      interpretations
                    </p>
                  </div>
                )}
              </div>
            </div>
          )}

          {/* Dashboard tab */}
          {activeTab === "dashboard" && (
            <div className="mb-8">
              <ExplainabilityDashboard assessment={assessment} />
```

```
            </div>
          )}

          {/* Trace tab */}
          {activeTab === "trace" && (
            <div className="mb-8">
              <ExecutionTrace trace={assessment.execution_trace} />
            </div>
          )}

          {/* Stage comparison always visible below tabs */}
          <section className="mb-8">
            <StageComparison assessment={assessment} />
          </section>
        </>
      )}
    </main>

    {/* Footer */}
    <footer className="border-t border-[var(--color-cyber-border)] py-4">
      <div className="max-w-7xl mx-auto px-6 flex items-center justify-between text-xs text-[var(--color-cyber-text-dim)]">
        <p>CyberExplain AI — College Mini-Project</p>
        <p>
          Powered by LangGraph + Groq + Next.js
        </p>
      </div>
    </footer>
  </div>
  );
}
```

File: Interactive Decision Graph (React Flow) ***(frontend/src/components/AgentGraph.tsx)***

```
"use client";

import { useCallback, useMemo } from "react";
import {
  ReactFlow,
  Controls,
  Background,
  BackgroundVariant,
  type Node,
  type Edge,
  type NodeTypes,
  Handle,
  Position,
} from "@xyflow/react";
import "@xyflow/react/dist/style.css";

import type {
  AssessmentResult,
  AgentNodeData,
  GraphEdge as GEdge,
} from "@/types";

/* ■■ Custom node component ■■ */

interface CustomNodeProps {
  data: AgentNodeData & { onSelect: (id: string) => void };
}

const stageColors: Record<string, { bg: string; border: string; glow: string }> = {
  input: {
    bg: "from-slate-600/20 to-slate-700/20",
    border: "border-slate-500/40",
    glow: "",
  },
  initial: {
    bg: "from-cyan-600/20 to-blue-700/20",
    border: "border-cyan-500/40",
    glow: "glow-cyan",
  },
  revision: {
    bg: "from-amber-600/20 to-orange-700/20",
    border: "border-amber-500/40",
    glow: "glow-orange",
  },
  final: {
```

```
    bg: "from-emerald-600/20 to-green-700/20",
    border: "border-emerald-500/40",
    glow: "glow-green",
  },
};

const stageIcons: Record<string, string> = {
  input: "■",
  initial: "■",
  revision: "■",
  final: "■",
};

function AgentNode({ data }: CustomNodeProps) {
  const colors = stageColors[data.stage] || stageColors.initial;
  const icon = stageIcons[data.stage] || "■";

  return (
    <div
      onClick={() => data.onSelect(data.id)}
      className={`
        cursor-pointer rounded-xl border ${colors.border} ${data.active ? colors.glow : ""}
        bg-gradient-to-br ${colors.bg} backdrop-blur-sm
        px-5 py-3.5 min-w-[180px] transition-all duration-300
        hover:scale-[1.03] hover:shadow-lg
        ${data.active ? "ring-1 ring-cyan-400/30" : ""}
      `}
    >
      <Handle
        type="target"
        position={Position.Top}
        className="!bg-[var(--color-cyber-accent)] !border-[var(--color-cyber-bg)] !w-2.5 !h-2.5"
      />

      <div className="flex items-center gap-2 mb-1.5">
        <span className="text-base">{icon}</span>
        <span className="font-semibold text-sm text-[var(--color-cyber-text)]">
          {data.label}
        </span>
      </div>

      {data.status && (
        <span className={`badge badge-${data.status.toLowerCase().replace(/ /g, "_")}`}>
          {data.status}
        </span>
```

```
      )}

      {data.confidence !== undefined && data.confidence > 0 && (
        <div className="mt-2">
          <div className="confidence-bar">
            <div
              className="confidence-bar-fill"
              style={{
                width: `${data.confidence * 100}%`,
                background:
                  data.confidence > 0.7
                    ? "var(--color-cyber-green)"
                    : data.confidence > 0.4
                    ? "var(--color-cyber-orange)"
                    : "var(--color-cyber-red)",
              }}
            />
          </div>
          <p className="text-[10px] text-[var(--color-cyber-text-dim)] mt-0.5">
            Confidence: {(data.confidence * 100).toFixed(0)}%
          </p>
        </div>
      )}

      <Handle
        type="source"
        position={Position.Bottom}
        className="!bg-[var(--color-cyber-accent)] !border-[var(--color-cyber-bg)] !w-2.5 !h-2.5"
      />
    </div>
  );
}

const nodeTypes: NodeTypes = { agentNode: AgentNode };

/* ■■ Build nodes & edges from assessment data ■■ */

interface Props {
  assessment: AssessmentResult;
  onNodeSelect: (nodeId: string) => void;
}

export default function AgentGraph({ assessment, onNodeSelect }: Props) {
  const onSelect = useCallback(
    (id: string) => onNodeSelect(id),
```

```
136 |     [onNodeSelect]
137 |   );
138 |
139 |   const revisionHappened =
140 |     assessment.final_assessment?.before_after?.revision_performed ?? false;
141 |
142 |   const nodes: Node[] = useMemo(() => {
143 |     const base: Node[] = [
144 |       {
145 |         id: "scenario_input",
146 |         type: "agentNode",
147 |         position: { x: 300, y: 0 },
148 |         data: {
149 |           id: "scenario_input",
150 |           label: "Scenario Input",
151 |           stage: "input",
152 |           active: true,
153 |           onSelect,
154 |         } as AgentNodeData & { onSelect: (id: string) => void },
155 |       },
156 |       {
157 |         id: "threat_agent",
158 |         type: "agentNode",
159 |         position: { x: 150, y: 120 },
160 |         data: {
161 |           id: "threat_agent",
162 |           label: "Threat Agent",
163 |           stage: "initial",
164 |           confidence:
165 |             assessment.initial_threat_analysis?.overall_confidence ?? 0,
166 |           status:
167 |             (assessment.initial_threat_analysis?.threats?.length ?? 0) > 0
168 |               ? `${assessment.initial_threat_analysis?.threats?.length} threats`
169 |               : undefined,
170 |           active: true,
171 |           onSelect,
172 |         } as AgentNodeData & { onSelect: (id: string) => void },
173 |       },
174 |       {
175 |         id: "risk_agent",
176 |         type: "agentNode",
177 |         position: { x: 450, y: 120 },
178 |         data: {
179 |           id: "risk_agent",
180 |           label: "Risk Agent",
```

```
          stage: "initial",
          confidence: assessment.initial_risk_assessment?.confidence ?? 0,
          status: assessment.initial_risk_assessment?.severity,
          active: true,
          onSelect,
        } as AgentNodeData & { onSelect: (id: string) => void },
      },
      {
        id: "verification_agent",
        type: "agentNode",
        position: { x: 300, y: 260 },
        data: {
          id: "verification_agent",
          label: "Verification Agent",
          stage: "initial",
          confidence:
            assessment.verification_result?.evidence_quality_score ?? 0,
          status: assessment.verification_result?.overall_status,
          active: true,
          onSelect,
        } as AgentNodeData & { onSelect: (id: string) => void },
      },
    ];

    if (revisionHappened) {
      base.push(
        {
          id: "threat_agent_revision",
          type: "agentNode",
          position: { x: 100, y: 400 },
          data: {
            id: "threat_agent_revision",
            label: "Threat Agent (Revised)",
            stage: "revision",
            confidence:
              assessment.revised_threat_analysis?.overall_confidence ?? 0,
            active: true,
            onSelect,
          } as AgentNodeData & { onSelect: (id: string) => void },
        },
        {
          id: "risk_agent_revision",
          type: "agentNode",
          position: { x: 500, y: 400 },
          data: {
```

```
226 |             id: "risk_agent_revision",
227 |             label: "Risk Agent (Revised)",
228 |             stage: "revision",
229 |             confidence:
230 |               assessment.revised_risk_assessment?.confidence ?? 0,
231 |             status: assessment.revised_risk_assessment?.severity,
232 |             active: true,
233 |             onSelect,
234 |           } as AgentNodeData & { onSelect: (id: string) => void },
235 |         }
236 |       );
237 |     }
238 |
239 |     base.push({
240 |       id: "final_assessment",
241 |       type: "agentNode",
242 |       position: { x: 300, y: revisionHappened ? 540 : 400 },
243 |       data: {
244 |         id: "final_assessment",
245 |         label: "Final Assessment",
246 |         stage: "final",
247 |         confidence: assessment.final_assessment?.confidence ?? 0,
248 |         status: assessment.final_assessment?.severity,
249 |         active: true,
250 |         onSelect,
251 |       } as AgentNodeData & { onSelect: (id: string) => void },
252 |     });
253 |
254 |     return base;
255 |   }, [assessment, revisionHappened, onSelect]);
256 |
257 |   const edges: Edge[] = useMemo(() => {
258 |     const base: Edge[] = [
259 |       {
260 |         id: "e-input-threat",
261 |         source: "scenario_input",
262 |         target: "threat_agent",
263 |         animated: true,
264 |         label: "scenario",
265 |         style: { stroke: "var(--color-cyber-accent)" },
266 |         labelStyle: { fill: "var(--color-cyber-text-dim)", fontSize: 10 },
267 |       },
268 |       {
269 |         id: "e-input-risk",
270 |         source: "scenario_input",
```

```
271 |         target: "risk_agent",
272 |         animated: true,
273 |         label: "scenario",
274 |         style: { stroke: "var(--color-cyber-accent)" },
275 |         labelStyle: { fill: "var(--color-cyber-text-dim)", fontSize: 10 },
276 |       },
277 |       {
278 |         id: "e-threat-verify",
279 |         source: "threat_agent",
280 |         target: "verification_agent",
281 |         animated: true,
282 |         label: "threats",
283 |         style: { stroke: "var(--color-cyber-blue)" },
284 |         labelStyle: { fill: "var(--color-cyber-text-dim)", fontSize: 10 },
285 |       },
286 |       {
287 |         id: "e-risk-verify",
288 |         source: "risk_agent",
289 |         target: "verification_agent",
290 |         animated: true,
291 |         label: "risk assessment",
292 |         style: { stroke: "var(--color-cyber-blue)" },
293 |         labelStyle: { fill: "var(--color-cyber-text-dim)", fontSize: 10 },
294 |       },
295 |     ];
296 |
297 |     if (revisionHappened) {
298 |       base.push(
299 |         {
300 |           id: "e-verify-threat-rev",
301 |           source: "verification_agent",
302 |           target: "threat_agent_revision",
303 |           animated: true,
304 |           label: "corrections",
305 |           style: { stroke: "var(--color-cyber-orange)" },
306 |           labelStyle: { fill: "var(--color-cyber-text-dim)", fontSize: 10 },
307 |         },
308 |         {
309 |           id: "e-verify-risk-rev",
310 |           source: "verification_agent",
311 |           target: "risk_agent_revision",
312 |           animated: true,
313 |           label: "corrections",
314 |           style: { stroke: "var(--color-cyber-orange)" },
315 |           labelStyle: { fill: "var(--color-cyber-text-dim)", fontSize: 10 },
```

```
        },
        {
          id: "e-threat-rev-final",
          source: "threat_agent_revision",
          target: "final_assessment",
          animated: true,
          label: "revised threats",
          style: { stroke: "var(--color-cyber-green)" },
          labelStyle: { fill: "var(--color-cyber-text-dim)", fontSize: 10 },
        },
        {
          id: "e-risk-rev-final",
          source: "risk_agent_revision",
          target: "final_assessment",
          animated: true,
          label: "revised risk",
          style: { stroke: "var(--color-cyber-green)" },
          labelStyle: { fill: "var(--color-cyber-text-dim)", fontSize: 10 },
        }
      );
    } else {
      base.push({
        id: "e-verify-final",
        source: "verification_agent",
        target: "final_assessment",
        animated: true,
        label: "verified",
        style: { stroke: "var(--color-cyber-green)" },
        labelStyle: { fill: "var(--color-cyber-text-dim)", fontSize: 10 },
      });
    }

    return base;
  }, [revisionHappened]);

  return (
    <div className="cyber-card p-0 overflow-hidden" style={{ height: revisionHappened ? 680 : 520 }}>
      <div className="px-5 py-3 border-b border-[var(--color-cyber-border)] flex items-center justify-between">
        <h3 className="font-semibold text-sm">
          ■ Agent Decision Graph
        </h3>
        <p className="text-xs text-[var(--color-cyber-text-dim)]">
          Click any node to inspect details
        </p>
      </div>
```

```
      <ReactFlow
        nodes={nodes}
        edges={edges}
        nodeTypes={nodeTypes}
        fitView
        fitViewOptions={{ padding: 0.3 }}
        proOptions={{ hideAttribution: true }}
        minZoom={0.5}
        maxZoom={1.5}
        nodesDraggable={true}
        nodesConnectable={false}
      >
        <Background variant={BackgroundVariant.Dots} gap={20} size={1} color="var(--color-cyber-border)" />
        <Controls showInteractive={false} />
      </ReactFlow>
    </div>
  );
}

```

File: Explainability Claims Dashboard ***(frontend/src/components/ExplainabilityDashboard.tsx)***

```
"use client";

import type { AssessmentResult } from "@/types";

interface Props {
  assessment: AssessmentResult;
}

export default function ExplainabilityDashboard({ assessment }: Props) {
  const { final_assessment: fa, verification_result: vr } = assessment;

  return (
    <div className="space-y-6">
      {/* Top-level risk overview */}
      <div className="cyber-card glow-cyan">
        <h3 className="font-semibold text-base mb-4">
          ■■ Risk Assessment Overview
        </h3>
        <div className="grid grid-cols-2 md:grid-cols-4 gap-4">
          <StatCard
            label="Severity"
            value={fa?.severity || "—"}
            color={severityColor(fa?.severity)}
          />
          <StatCard
            label="Risk Score"
            value={`${fa?.risk_score || 0}/10`}
            color={scoreColor(fa?.risk_score)}
          />
          <StatCard
            label="Confidence"
            value={`${((fa?.confidence || 0) * 100).toFixed(0)}%`}
            color={confidenceColor(fa?.confidence)}
          />
          <StatCard
            label="Verification"
            value={fa?.verification_status || "—"}
            color={
              fa?.verification_status === "verified"
                ? "text-emerald-400"
                : "text-amber-400"
            }
          />
        </div>
      </div>
```

```
46 |
47 |       {/* Verification Claims Breakdown */}
48 |       {vr && (
49 |         <div className="cyber-card">
50 |           <h3 className="font-semibold text-base mb-4">
51 |             ■ Verification Breakdown
52 |           </h3>
53 |
54 |           {/* Stats bar */}
55 |           <div className="flex gap-4 mb-4">
56 |             <MiniStat
57 |               label="Supported"
58 |               count={vr.supported_count ?? 0}
59 |               color="bg-emerald-500"
60 |             />
61 |             <MiniStat
62 |               label="Unsupported"
63 |               count={vr.unsupported_count ?? 0}
64 |               color="bg-amber-500"
65 |             />
66 |             <MiniStat
67 |               label="Contradicted"
68 |               count={vr.contradicted_count ?? 0}
69 |               color="bg-red-500"
70 |             />
71 |           </div>
72 |
73 |           {/* Evidence quality */}
74 |           {vr.evidence_quality_score !== undefined && (
75 |             <div className="mb-4">
76 |               <p className="text-xs text-[var(--color-cyber-text-dim)] mb-1">
77 |                 Evidence Quality Score
78 |               </p>
79 |               <div className="confidence-bar">
80 |                 <div
81 |                   className="confidence-bar-fill"
82 |                   style={{
83 |                     width: `${vr.evidence_quality_score * 100}%`,
84 |                     background:
85 |                       vr.evidence_quality_score > 0.7
86 |                         ? "var(--color-cyber-green)"
87 |                         : "var(--color-cyber-orange)",
88 |                   }}
89 |                 />
90 |               </div>
```

```
              <p className="text-xs text-[var(--color-cyber-text-dim)] mt-0.5">
                {(vr.evidence_quality_score * 100).toFixed(0)}%
              </p>
            </div>
          )}

          {/* Supported */}
          {(vr.supported_claims?.length ?? 0) > 0 && (
            <ClaimSection
              title="Supported Claims"
              icon="■"
              claims={vr.supported_claims!}
              color="emerald"
            />
          )}

          {/* Unsupported */}
          {(vr.unsupported_claims?.length ?? 0) > 0 && (
            <ClaimSection
              title="Unsupported Claims"
              icon="■■"
              claims={vr.unsupported_claims!}
              color="amber"
            />
          )}

          {/* Contradictions */}
          {(vr.contradictions?.length ?? 0) > 0 && (
            <ClaimSection
              title="Contradictions"
              icon="■"
              claims={vr.contradictions!}
              color="red"
            />
          )}

          {/* Disagreements */}
          {(vr.disagreements?.length ?? 0) > 0 && (
            <div className="mt-4">
              <h4 className="text-xs font-semibold text-[var(--color-cyber-text-dim)] uppercase tracking-wider mb-2">
                ■■ Agent Disagreements
              </h4>
              <div className="space-y-2">
                {vr.disagreements!.map((d, i) => (
                  <div
```

```
                    key={i}
                    className="bg-purple-500/5 border border-purple-500/20 rounded-lg p-3"
                  >
                    <p className="text-sm font-medium text-purple-400 mb-1">
                      {d.topic}
                    </p>
                    <p className="text-xs text-[var(--color-cyber-text-dim)]">
                      <strong>Threat Agent:</strong> {d.threat_agent_position}
                    </p>
                    <p className="text-xs text-[var(--color-cyber-text-dim)]">
                      <strong>Risk Agent:</strong> {d.risk_agent_position}
                    </p>
                    <p className="text-xs text-emerald-400 mt-1">
                      <strong>Evidence:</strong> {d.evidence_assessment}
                    </p>
                  </div>
                ))}
              </div>
            </div>
          )}

          {/* Corrections */}
          {(vr.corrections_requested?.length ?? 0) > 0 && (
            <div className="mt-4">
              <h4 className="text-xs font-semibold text-[var(--color-cyber-text-dim)] uppercase tracking-wider mb-2">
                ■ Corrections Requested
              </h4>
              <ul className="space-y-1">
                {vr.corrections_requested!.map((c, i) => (
                  <li
                    key={i}
                    className="text-sm bg-amber-500/5 border border-amber-500/20 rounded-lg px-3 py-2 text-amber-300"
                  >
                    {c}
                  </li>
                ))}
              </ul>
            </div>
          )}
        </div>
      )}

      {/* Recommendations */}
      {(fa?.recommendations?.length ?? 0) > 0 && (
        <div className="cyber-card">
```

```
          <h3 className="font-semibold text-base mb-3">
            ■ Recommendations
          </h3>
          <ul className="space-y-2">
            {fa!.recommendations.map((r, i) => (
              <li
                key={i}
                className="flex items-start gap-2 text-sm text-[var(--color-cyber-text)]"
              >
                <span className="text-[var(--color-cyber-accent)] mt-0.5 shrink-0">
                  →
                </span>
                {r}
              </li>
            ))}
          </ul>
        </div>
      )}

      {/* Disclaimer */}
      {fa?.disclaimer && (
        <div className="rounded-lg border border-amber-500/20 bg-amber-500/5 p-4">
          <p className="text-xs text-amber-400/80 flex items-start gap-2">
            <span className="shrink-0 mt-0.5">■■</span>
            {fa.disclaimer}
          </p>
        </div>
      )}
    </div>
  );
}

/* ■■ Sub-components ■■ */

function StatCard({
  label,
  value,
  color,
}: {
  label: string;
  value: string;
  color: string;
}) {
  return (
    <div className="bg-[var(--color-cyber-bg)] rounded-xl p-4 border border-[var(--color-cyber-border)]">
```

```
226 |       <p className="text-xs text-[var(--color-cyber-text-dim)] mb-1">{label}</p>
227 |       <p className={`text-xl font-bold ${color}`}>{value}</p>
228 |     </div>
229 |   );
230 | }
231 |
232 | function MiniStat({
233 |   label,
234 |   count,
235 |   color,
236 | }: {
237 |   label: string;
238 |   count: number;
239 |   color: string;
240 | }) {
241 |   return (
242 |     <div className="flex items-center gap-2">
243 |       <span className={`w-2.5 h-2.5 rounded-full ${color}`} />
244 |       <span className="text-sm text-[var(--color-cyber-text)]">
245 |         {count} {label}
246 |       </span>
247 |     </div>
248 |   );
249 | }
250 |
251 | function ClaimSection({
252 |   title,
253 |   icon,
254 |   claims,
255 |   color,
256 | }: {
257 |   title: string;
258 |   icon: string;
259 |   claims: { claim: string; explanation: string; evidence?: string; missing_evidence?: string }[];
260 |   color: string;
261 | }) {
262 |   return (
263 |     <div className="mt-4">
264 |       <h4 className="text-xs font-semibold text-[var(--color-cyber-text-dim)] uppercase tracking-wider mb-2">
265 |         {icon} {title}
266 |       </h4>
267 |       <div className="space-y-2">
268 |         {claims.map((c, i) => (
269 |           <div
270 |             key={i}
```

```
            className={`bg-${color}-500/5 border border-${color}-500/20 rounded-lg p-3`}
          >
            <p className={`text-sm font-medium text-${color}-400`}>
              {c.claim}
            </p>
            <p className="text-xs text-[var(--color-cyber-text-dim)] mt-1">
              {c.explanation}
            </p>
            {c.evidence && (
              <p className="text-xs text-[var(--color-cyber-text-dim)] mt-0.5">
                <span className="font-medium">Evidence:</span> {c.evidence}
              </p>
            )}
            {c.missing_evidence && (
              <p className="text-xs text-amber-400/70 mt-0.5">
                <span className="font-medium">Missing:</span>{" "}
                {c.missing_evidence}
              </p>
            )}
          </div>
        ))}
      </div>
    </div>
  );
}

/* ■■ Helpers ■■ */

function severityColor(s?: string): string {
  switch (s?.toLowerCase()) {
    case "critical":
      return "text-red-400";
    case "high":
      return "text-amber-400";
    case "medium":
      return "text-blue-400";
    case "low":
      return "text-emerald-400";
    default:
      return "text-[var(--color-cyber-text)]";
  }
}

function scoreColor(s?: number): string {
  if (!s) return "text-[var(--color-cyber-text)]";
  if (s >= 8) return "text-red-400";
  if (s >= 6) return "text-amber-400";
  if (s >= 4) return "text-blue-400";
  return "text-emerald-400";
}

function confidenceColor(c?: number): string {
  if (!c) return "text-[var(--color-cyber-text)]";
  if (c >= 0.7) return "text-emerald-400";
  if (c >= 0.4) return "text-amber-400";
  return "text-red-400";
}

```

File: Stage 1 vs Stage 2 Comparison ***(frontend/src/components/StageComparison.tsx)***

```
"use client";

import type { AssessmentResult } from "@/types";

interface Props {
  assessment: AssessmentResult;
}

export default function StageComparison({ assessment }: Props) {
  const ba = assessment.final_assessment?.before_after;
  if (!ba) return null;

  const severityChanged = ba.initial_severity !== ba.final_severity;
  const scoreChanged = ba.initial_risk_score !== ba.final_risk_score;
  const confidenceChanged = ba.initial_confidence !== ba.final_confidence;
  const threatCountChanged = ba.initial_threat_count !== ba.final_threat_count;

  return (
    <div className="cyber-card">
      <h3 className="font-semibold text-base mb-1">
        ■ Before vs After Verification
      </h3>
      <p className="text-xs text-[var(--color-cyber-text-dim)] mb-4">
        Comparing initial assessment (Stage 1) with revised assessment (Stage 2)
        {!ba.revision_performed && " — No revision was needed"}
      </p>

      <div className="grid grid-cols-1 md:grid-cols-2 gap-4">
        {/* Stage 1 */}
        <div className="rounded-xl border border-cyan-500/20 bg-cyan-500/5 p-4">
          <h4 className="font-semibold text-sm text-cyan-400 mb-3 flex items-center gap-2">
            <span className="w-2 h-2 rounded-full bg-cyan-500" />
            Stage 1 — Initial Assessment
          </h4>
          <MetricRow label="Severity" value={ba.initial_severity} badge />
          <MetricRow
            label="Risk Score"
            value={`${ba.initial_risk_score}/10`}
          />
          <MetricRow
            label="Confidence"
            value={`${(ba.initial_confidence * 100).toFixed(0)}%`}
          />
          <MetricRow label="Threats" value={`${ba.initial_threat_count}`} />
          <MetricRow
```

```
46 |             label="Unsupported Claims"
47 |             value={`${ba.unsupported_claims_found}`}
48 |             highlight={ba.unsupported_claims_found > 0}
49 |           />
50 |         </div>
51 |
52 |         {/* Stage 2 */}
53 |         <div className="rounded-xl border border-emerald-500/20 bg-emerald-500/5 p-4">
54 |           <h4 className="font-semibold text-sm text-emerald-400 mb-3 flex items-center gap-2">
55 |             <span className="w-2 h-2 rounded-full bg-emerald-500" />
56 |             Stage 2 — After Verification
57 |           </h4>
58 |           <MetricRow
59 |             label="Severity"
60 |             value={ba.final_severity}
61 |             badge
62 |             changed={severityChanged}
63 |           />
64 |           <MetricRow
65 |             label="Risk Score"
66 |             value={`${ba.final_risk_score}/10`}
67 |             changed={scoreChanged}
68 |           />
69 |           <MetricRow
70 |             label="Confidence"
71 |             value={`${(ba.final_confidence * 100).toFixed(0)}%`}
72 |             changed={confidenceChanged}
73 |           />
74 |           <MetricRow
75 |             label="Threats"
76 |             value={`${ba.final_threat_count}`}
77 |             changed={threatCountChanged}
78 |           />
79 |           <MetricRow
80 |             label="Revision Performed"
81 |             value={ba.revision_performed ? "Yes" : "No"}
82 |           />
83 |         </div>
84 |       </div>
85 |
86 |       {/* Change summary */}
87 |       {ba.revision_performed && (
88 |         <div className="mt-4 p-3 rounded-lg bg-amber-500/5 border border-amber-500/20">
89 |           <h4 className="text-xs font-semibold text-amber-400 uppercase tracking-wider mb-2">
90 |             What Changed
```

```
          </h4>
          <ul className="space-y-1 text-sm text-[var(--color-cyber-text-dim)]">
            {severityChanged && (
              <li className="changed-highlight">
                Severity: <span className="text-cyan-400">{ba.initial_severity}</span> →{" "}
                <span className="text-emerald-400">{ba.final_severity}</span>
              </li>
            )}
            {scoreChanged && (
              <li className="changed-highlight">
                Risk Score: <span className="text-cyan-400">{ba.initial_risk_score}</span> →{" "}
                <span className="text-emerald-400">{ba.final_risk_score}</span>
              </li>
            )}
            {confidenceChanged && (
              <li className="changed-highlight">
                Confidence: <span className="text-cyan-400">{(ba.initial_confidence * 100).toFixed(0)}%</span> →{" "}
                <span className="text-emerald-400">{(ba.final_confidence * 100).toFixed(0)}%</span>
              </li>
            )}
            {threatCountChanged && (
              <li className="changed-highlight">
                Threat count: <span className="text-cyan-400">{ba.initial_threat_count}</span> →{" "}
                <span className="text-emerald-400">{ba.final_threat_count}</span>
              </li>
            )}
            {ba.unsupported_claims_found > 0 && (
              <li className="changed-highlight">
                <span className="text-amber-400">{ba.unsupported_claims_found}</span> unsupported
                claim(s) identified and addressed
              </li>
            )}
          </ul>
        </div>
      )}
    </div>
  );
}

function MetricRow({
  label,
  value,
  badge = false,
  changed = false,
  highlight = false,
```

```
}: {
  label: string;
  value: string;
  badge?: boolean;
  changed?: boolean;
  highlight?: boolean;
}) {
  return (
    <div
      className={`flex items-center justify-between py-1.5 ${
        changed ? "changed-highlight rounded" : ""
      }`}
    >
      <span className="text-xs text-[var(--color-cyber-text-dim)]">{label}</span>
      {badge ? (
        <span className={`badge badge-${value.toLowerCase()}`}>{value}</span>
      ) : (
        <span
          className={`text-sm font-medium ${
            highlight ? "text-amber-400" : "text-[var(--color-cyber-text)]"
          } ${changed ? "text-emerald-400" : ""}`}
        >
          {value}
        </span>
      )}
    </div>
  );
}

```

7. Setup, Deployment & Verification Manual

1. GitHub Repository & Prerequisites:
• GitHub Repository: https://github.com/umar-khatri/cyber-explain-agent.git
• Python ≥ 3.10 (Tested on Python 3.13 macOS ARM64)
• Node.js ≥ 18 (Tested with Next.js 16.4 Turbopack)
• Groq API Key (using free tier openai/gpt-oss-120b)
• Supabase Account (with executed backend/schema.sql)

2. Backend Setup:
cd backend python3 -m venv venv source venv/bin/activate pip install -r requirements.txt cp .env.example .env # Populate GROQ_API_KEY, SUPABASE_URL, and SUPABASE_KEY in .env uvicorn app.main:app --reload --port 8000 API documentation available at: http://localhost:8000/docs

3. Frontend Setup:
cd frontend npm install # Ensure .env.local has NEXT_PUBLIC_API_URL=http://localhost:8000 npm run dev Web application available at: http://localhost:3000

4. End-to-End Test Execution:
The backend test suite validates all endpoints and simulates the multi-agent decision cycle: python3 -m py_compile backend/app/*.py backend/app/agents/*.py backend/app/routes/*.py cd frontend && npm run build

8. Conclusion & Future Directions

CyberExplain AI successfully demonstrates that multi-agent architectures provide the essential checks and balances required for enterprise AI adoption in cybersecurity. By disaggregating threat detection from severity rating and introducing an explicit verification agent that challenges ungrounded assertions, organizations can prevent costly hallucinated incident panics while maintaining complete auditability through transparent execution traces.